SAHIL SAHRAN

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EDUCATION

University of Toronto **Toronto, ON**

Bachelor of Applied Science in Electrical Engineering + PEY Co-op *Sep 2025 – Apr 2030 (expected)*

- Ranked among the top 30 students in the ECE cohort with Dean's List standing, maintaining a 3.97/4.00 cumulative GPA.
- Completed coursework in Introductory Electronics covering diodes, transistors, and op-amps, Digital Systems covering Verilog and FPGA design, and Programming Fundamentals covering C++.

EXPERIENCE

Embedded Systems Intern **Jun 2026 – Aug 2026**

Nanotronic Electronic *North York, ON*

- Designed circuit schematics in Altium Designer for an RC aircraft transmitter combining an ESP32, RYLR993 LoRa transceiver, and a USB-to-SPI joystick bridge. Modeled all component footprints and schematic symbols from scratch, adding verified JLCPCB part numbers to ensure direct BOM ordering.
- Evaluated NAU7802 versus HX711 24-bit ADCs for a weight dispenser, wiring a load cell and writing prototype sensor firmware in Arduino. Proposed a dedicated LDO regulator that independently halved supply noise, then relocated decoupling capacitors directly across the ADC pins and enabled the unused channel's PGA filter capacitor to drop total noise from ± 1 g down to ± 50 mg.
- Populated 100+ medical project boards combining stencil paste reflow with hand soldering. Bench-tested every completed assembly, troubleshooting and repairing shorted power rails, cold joints, and reversed diodes.

CNC Programmer **May 2026 – Sep 2026**

D2Woodworking *Markham, ON*

- Ran shop CNC jobs independently across 3 whole-house builds and 5+ smaller custom projects totaling over $100K CAD. Programmed toolpaths in Mozaik CAD/CAM directly from architectural plans, where a single machining error meant scrapped stock, shop rework, or delayed installations.

PROJECTS

ESPRA: Always-On Desk Assistant Robot | ESP32-S3, C++, FreeRTOS, Python **Jun 2026 – Oct 2026**

- Built a differential-drive robot with voice control on an ESP32-S3, powering N20 gear motors through a TB6612FNG driver. Regulated motion using a 50 Hz FreeRTOS task with gradual acceleration ramps and a time-of-flight collision interlock at 50 mm.
- Diagnosed spontaneous microcontroller reboots by A/B testing the ESP32 on PC USB versus a shared power-bank rail. Resolved brownout resets by installing bulk and decoupling capacitors, adding pull-up resistors on the I2C bus, and soft-starting the motors in firmware.
- Enabled the robot to pivot toward speakers by executing acoustic time-difference-of-arrival (TDOA) angle estimation on the ESP32 using a pair of microphones spaced 8.25 cm apart.

Beehive Winter Monitoring System | Client: UofT BEES, team of 6 **Jan 2026 – Apr 2026**

- Implemented sensor power gating using an AO3401A P-MOSFET high-side switch with an RC soft-start circuit, a 1N5817 reverse-polarity Schottky diode, and a battery-sense divider. Restricted average draw below 0.6 mA by waking every 30 minutes to measure before returning to deep sleep, alongside hourly Google Sheets HTTPS syncs. This low average current was calculated to support an estimated 150-day winter lifespan on $3\times$ lithium AA cells.
- Calculated an 11.2 cm minimum safe distance from the ESP32 transmitter using free-space power density and 377 Ω wave impedance. Kept RF exposure below the 1.94 V/m limit from published bee stress studies and confirmed field strength with an EMF meter at $\sim$1 V/m.

Size-Targeted JPEG Compression | Independent Research, MATLAB **Aug 2024 – Nov 2024**

- Modified JPEG to target a user-specified file size rather than a quality level. Solved for the zero-rounding threshold and quantization table scale using Lagrange multipliers, minimizing mean squared error.
- Formulated an original predictive model for file size by fitting a Laplacian distribution to DCT coefficients and computing average Huffman code lengths from ITU-T T.81 standards. Predicted compressed output sizes within a 5.6% mean error across 5 benchmark photographs.

SKILLS

Programming: C, C++, Python (NumPy), MATLAB, Verilog
Hardware: Altium Designer, LTspice, Quartus Prime, soldering (through-hole, SMD stencil/reflow), board-level debugging
Embedded: ESP32 (Arduino framework), FreeRTOS, I2C, I2S, SPI, UART, 1-Wire, DMA audio streaming, Wi-Fi web servers
CAD & Languages: Autodesk Fusion, SketchUp, Mozaik CAD/CAM | English (native), French (working proficiency)